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VU Answer

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An ordered pair corresponds to ----- on the plane

Answer (Please select your correct option)

☐ A unique point

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☐ A point in each quadrant

☐ Two points

☐ Infinite number of points

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Plane is a special type of -----.

Answer (Please select your correct option)

☐ Curve

☐ Surface

☐ Sphere

☐ Cone

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There is one-to-one correspondence between the set of points on a co-ordinate line and -----

Answer (Please select your correct option)



Set of real numbers



Set of integers



Set of natural numbers



Set of rational numbers

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What is the general equation of parabola whose axis of symmetry is parallel to y-axis?

Answer (Please select your correct option)

☐

$$y = ax^2 + b \quad (a \neq 0)$$

☐

$$x = ay^2 + b \quad (a \neq 0)$$

☐

$$y = ax^2 + bx + c \quad (a \neq 0)$$

☐

$$x = ay^2 + by + c \quad (a \neq 0)$$

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If the spherical coordinates of a point are $\left(2, \frac{\pi}{4}, 0\right)$, then z-coordinate in rectangular coordinate system is

Answer (Please select your correct option)

☐ 0☐ -2☐ 2☐ $\sqrt{2}$

The function $z = \frac{1}{\sqrt{x+y}}$ is discontinuous at origin because at the point (0,0), it

Answer (Please select your correct option)

- ☐ has a jump
- ☐ has a hole
- ☐ approaches towards infinity
- ☐ approaches towards zero

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The function $f(x, y) = \sqrt{y - x}$ is continuous in the region ----- and discontinuous elsewhere.

Answer (Please select your correct option)

☐ $x \geq y$

☐ $x \neq y$

☐ $x \leq y$

☐ $x > y$

Let $w = f(x, y, z)$ and $x = g(r, s)$, $y = h(r, s)$, $z = t(r, s)$ then by chain rule $\frac{\partial w}{\partial r} = \dots\dots\dots$

Answer (Please select your correct option)

☐ $\frac{\partial w}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial r}$

☐ $\frac{\partial w}{\partial r} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial r} \frac{\partial y}{\partial r} + \frac{\partial w}{\partial r} \frac{\partial z}{\partial r}$

☐ $\frac{\partial w}{\partial x} \frac{\partial x}{\partial r} \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r} \frac{\partial y}{\partial s} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial r} \frac{\partial z}{\partial s}$

☐ $\frac{\partial w}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial w}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial w}{\partial r} \frac{\partial r}{\partial z}$

Every differentiable function is always

Answer (Please select your correct option)

☐ Piece wise continuous

☐ Discontinuous

☐ Continuous

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The vector $\vec{a} \times \vec{b}$ is to both $\vec{a}$ and $\vec{b}$.

Answer (Please select your correct option)

☐ Opposite

☐ Equal

☐ Parallel

☐ Orthogonal

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Gradient of a scalar function always results in a function.

Answer (Please select your correct option)

☐

Scalar

☐

Continuous

☐

Vector

☐

Constant

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The direction of gradient at any point on the surface is to the tangent plane at that point.

Answer (Please select your correct option)

- ☐ parallel
- ☐ perpendicular
- ☐ opposite direction
- ☐ None of these.

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$f_x(p_0)(x - x_0) + f_y(p_0)(y - y_0) + f_z(p_0)(z - z_0) = 0$ is the equation of

Answer (Please select your correct option)

☐

Normal plane

☐

Point slope form of line

☐

Gradient of function $f(x, y, z)$

☐

Tangent plane

correct

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For a function $z = f(x, y)$, the total differential is defined as

Answer (Please select your correct option)

☐

$$dz = f_x(x, y)dx - f_y(x, y)dy$$

☐

$$dz = f_x(x, y)dx + f_y(x, y)dy$$

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☐

$$dz = f_x(x, y)dy - f_y(x, y)dx$$

☐

$$dz = f_x(x, y)dy + f_y(x, y)dx$$

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For a function $f(x,y)$ to have both absolute maximum and minimum, it must be Continuous on set R .

Answer (Please select your correct option)

☐ a closed and bounded

correct

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☐ an open and bounded

☐ a closed and unbounded

☐ an open and unbounded

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The volume of parallelepiped with dimensions x, y, z is

Answer (Please select your correct option)

☐

$$V = x^2 y^2 z^2$$

☐

$$V = x + y + z$$

☐

$$V = \sqrt{xyz}$$

☐

$$V = xyz$$

correct

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Let x be the length, width and height of a cube. The area of bottom will be

Answer (Please select your correct option)

☐

$$A = x^2$$

☐

$$A = x^3$$

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☐

$$A = xz$$

☐

$$A = x + x^2$$

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A double integral and iterated integral become identical provided that the integrand is over the given rectangular region.

Answer (Please select your correct option)

☐ Bounded

correct

☐ Discontinuous

☐ Continuous

☐ Defined

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If $R = \{(x, y) : 2 \leq x \leq 4 \text{ and } 0 \leq y \leq 1\}$, then $\iint_R (4xe^{2y}) dA = \dots\dots\dots$

Answer (Please select your correct option)

☐ $\int_0^1 \int_2^4 (4xe^{2y}) dy dx$

☐ $\int_0^1 \int_2^4 (4xe^{2y}) dx dy$

☐ $\int_1^4 \int_0^2 (4xe^{2y}) dx dy$

☐ $\int_1^4 \int_0^2 (4xe^{2y}) dy dx$

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If $R = \{(x, y) : 0 \leq x \leq 2 \text{ and } 0 \leq y \leq 3\}$, then $\iint_R (1 - ye^{xy}) dA = \dots\dots\dots$

Answer (Please select your correct option)

☐ $\int_0^2 \int_0^3 (1 - ye^{xy}) dy dx$

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☐ $\int_0^2 \int_0^3 (1 - ye^{xy}) dx dy$

☐ $\int_2^3 \int_0^0 (1 - ye^{xy}) dx dy$

☐ $\int_0^2 \int_2^3 (4xe^{2y}) dy dx$

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Which of the following is geometrical representation of set of real numbers?

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Answer (Please select your correct option)



Co-ordinate line



xy-plane

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Sphere



Circular cylinder

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There is one-to-one correspondence between the set of points on a co-ordinate line and -----

Answer (Please select your correct option)



Set of real numbers

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Set of integers



Set of natural numbers



Set of rational numbers

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An ordered triple corresponds to ----- in a three dimensional space.

Answer (Please select your correct option)



A unique point



A point in each octant



Three points



Infinite number of points

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If the positive directions of x and y axes are known then ----- the positive direction of z-axis.

Answer (Please select your correct option)



Horizontal rightward direction is



Horizontal leftward direction is



Left hand rule tells



Right hand rule tells

correct not shoure

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If a function is not defined at some point, then its limit ----- exist at that point.

Answer (Please select your correct option)

Always



Never



May



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A composition of continuous functions

Answer (Please select your correct option)



is always continuous

correct

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may or may not be continuous



is discontinuous



is piece-wise continuous

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According to the Euler's theorem, the order of partial differentiation can be changed, provided the function and all of its partial derivatives are.....

Answer (Please select your correct option)

☐ Piece-wise continuous

☐ Continuous

☐ Discontinuous

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If $x = f(r, s)$ and $r = g(t)$, $s = h(t)$, then derivative of x with respect to t is written as.....

Answer (Please select your correct option)

☐ $\frac{\partial x}{\partial t}$

☐ (x, t)

☐ $\frac{\partial^2 x}{\partial t^2}$

☐ $\frac{dx}{dt}$

correct

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A vector in a plane is always represented by a

Answer (Please select your correct option)

☐

Line

☐

Curve

☐

Line segment

☐

None of these.

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If two vectors $\vec{a}$ and $\vec{b}$ have the relation $\vec{a} = \lambda \vec{b}$ where λ is a none-zero scalar then $\vec{a}$ and $\vec{b}$ are to each other.

Answer (Please select your correct option)

☐ Orthogonal

☐ Non-parallel

☐ Parallel

☐ None of these



The function decreases most rapidly in the direction of

Answer (Please select your correct option)



$$-\nabla f$$



$$-|\nabla f|$$



$$\nabla f \times \hat{u}$$



$$|\nabla f|$$



The direction of gradient at any point on the surface is to the tangent plane at that point.

Answer (Please select your correct option)



parallel



perpendicular



opposite direction



None of these.

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$2x^2 + y^2 = 4$ is the form of equation of a curve.

Answer (Please select your correct option)



Parametric



Implicit

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Symmetric



Explicit

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If $f(x, y)$ has a relative extremum at a point (x_0, y_0) and both the first partial derivatives of f exist at this point, then

Answer (Please select your correct option)

☐

$$f_x(x_0, y_0) = 0 \text{ and } f_y(x_0, y_0) \neq 0$$

☐

$$f_x(x_0, y_0) \leq 0 \text{ and } f_y(x_0, y_0) \leq 0$$

☐

$$f_x(x_0, y_0) \geq 0 \text{ and } f_y(x_0, y_0) \geq 0$$

☐

$$f_x(x_0, y_0) = 0 \text{ and } f_y(x_0, y_0) = 0$$

correct

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For a function $f(x)$, if $f'(x)$ is equal to some non-zero constant, then $f(x)$ will have



Answer (Please select your correct option)



At least one critical point

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No critical point



Non-zero critical points



None of these

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Let x, y, z be the length, width and height of an open rectangular box. The surface area of the box will be

Answer (Please select your correct option)

☐

$$A = xy + 2yz + 2xz$$

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☐

$$A = yz + 4$$

☐

$$A = xz + yz + zx$$

☐

$$A = xyz$$

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Double integral of a function $f(x,y)$ represents of the region between the surface defined by the function and the plane which contains its domain.

Answer (Please select your correct option)



Perimeter



Volume



Area



Circumference

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For the double integral $\int_c^d \int_a^b f(x,y) dx dy$, order of integration does not matter provided that $f(x,y)$ is

Answer (Please select your correct option)



Bounded



Discontinuous



Defined



Continuous

If $R = \{(x, y) : 0 \leq x \leq 2 \text{ and } -1 \leq y \leq 1\}$, then $\iint_R (x + 2y^2) dA = \dots\dots\dots$

Answer (Please select your correct option)

☐ $\int_{-1}^1 \int_0^2 (x + 2y^2) dy dx$

☐ $\int_0^2 \int_1^{-1} (x + 2y^2) dx dy$

☐ $\int_{-1}^1 \int_0^2 (x + 2y^2) dx dy$

☐ $\int_1^2 \int_{-1}^0 (x + 2y^2) dx dy$

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If $R = \{(x, y) : 0 \leq x \leq 4 \text{ and } 0 \leq y \leq 9\}$, then $\iint_R (3x - 4x\sqrt{xy}) dA = \dots\dots\dots$

Answer (Please select your correct option)

☐ $\int_0^9 \int_0^4 (3x - 4x\sqrt{xy}) dy dx$

☐ $\int_0^4 \int_4^9 (3x - 4x\sqrt{xy}) dx dy$

☐ $\int_4^9 \int_0^4 (3x - 4x\sqrt{xy}) dx dy$

☐ $\int_0^4 \int_0^9 (3x - 4x\sqrt{xy}) dy dx$

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